Supplementary Material for: PhysioCAT: Physiological Delay-Banded Cross-Attention for Subject-Independent Cuffless Blood Pressure Estimation from ECG and PPG

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Supplementary Material

This supplement contains cohort definitions, fixed-protocol checks, mechanism controls, transfer analyses, subgroup results, and computational profiles supporting the main manuscript. Reference numbers follow the main manuscript.

S1. SQI definition, dataset retention, and reproducibility

Source-field and candidate lineage. The raw-source adapters support the public MATLAB layouts used by PulseDB and MIMIC-BP. For PulseDB, paired unfiltered `ECG_Record/PPG_Record` fields are preferred, with `ECG_Raw/PPG_Raw` as fallbacks. Each source-listed row yields at most one centered 8 s crop; crop and target-formation outcomes are retained in the public reproducibility repository.

Signal-quality index. Window-level SQI was computed separately for ECG and PPG, then used in the fixed retention rule and as a token-level reliability cue in PhysioCAT. Let $c(z) = \min(1, \max(0, z))$, $\epsilon_E = 10^{-8}$, and $\epsilon_P = 10^{-6}$. For an ECG window x_E with detected R peaks, define $E_E = \sum_t x_E(t)^2$, let E_R be the energy within ± 50 ms of the detected R peaks, and let κ_E denote Fisher kurtosis. The ECG SQI is

$$SQI_E = c\left\{\frac{1}{2}c\left(\frac{\kappa_E + 3}{20}\right) + \frac{1}{2}\frac{E_R}{E_E + \epsilon_E}\right\}. \tag{1}$$

For a median-centered PPG window x_P, let $X_P(f)$ denote its discrete Fourier transform and define

$$R_P = \frac{\sum_{0.5 \le f \le 4} |X_P(f)|^2}{\sum_{0.3 \le f \le 8} |X_P(f)|^2 + \epsilon_P}, \tag{2}$$

$$s_P = c\left\{\min\left(\frac{d_+}{d_-}, \frac{d_-}{d_+}\right)\right\}, \tag{3}$$

where $d_+ = \max(P_{99}(\nabla x_P), \epsilon_P)$ and $d_- = \max(|P_1(\nabla x_P)|, \epsilon_P)$. The PPG SQI is

$$SQI_P = c\left\{0.6 R_P + 0.4 s_P\right\}. \tag{4}$$

Both terms are invariant to affine amplitude scaling and therefore serve as morphology-quality cues rather than perfusion-amplitude features. All reported models receive only scalar SBP and DBP targets. For PulseDB-Vital and PulseDB-MIMIC, consecutive detected cardiac boundaries at least 280 ms apart defined candidate ABP beats. Non-finite or unordered extrema were removed, followed by a within-window $3.5 \times 1.4826 \times \mathrm{MAD}$ rule applied jointly to beat midpoint and pulse pressure; at least three accepted beats were required. The target was the mean of the accepted beat maxima and minima from the same synchronized 8 s crop. Cohort eligibility after target formation used the ECG/PPG checks and joint SQI rule below; reference-ABP quality is examined separately in Supplementary Table S28. MIMIC-BP contributes curated scalar labels paired with the released 8 s record. The fixed retention rule kept a window when

$$\min(SQI_E, SQI_P) \geq 0.40 \quad \text{and} \quad \max(SQI_E, SQI_P) \geq 0.55. \tag{5}$$

Supplementary Table S1 lists how the components enter preprocessing and token-level reliability cues.

Table S1: Signal-quality-index components used for preprocessing and token-level reliability cues.

Component	Definition	Physiological meaning
ECG SQI	Kurtosis term plus R-peak-neighborhood energy ratio.	Rewards sharp, repeatable QRS-dominant morphology.
PPG SQI	Pulse-band spectral concentration plus robust positive-versus-negative slope balance.	Rewards periodic pulse morphology without using absolute perfusion amplitude.
Token SQI	One-second sliding-window SQI values, advanced every 4 ms, interpolated to the centers of the 125 non-overlapping patches in each 8 s window.	Provides local reliability weights during fusion.

Table S2: Stepwise preprocessing retention for the primary PulseDB-Vital cohort.

Preprocessing step	Retained subjects	Retained windows	Window retention rate
Source-indexed adult waveform segments	2,714	240,428	100.0%
Paired ECG/PPG crop available	2,714	235,067	97.8%
Scalar target formed or paired	2,714	225,536	93.8%
At least six cardiac cycles	2,714	222,773	92.7%
ECG channel quality	2,714	216,078	89.9%
PPG channel quality	2,714	207,244	86.2%
Common paired-sample continuity	2,714	205,138	85.3%
Joint ECG/PPG SQI rule	2,714	193,533	80.5%
Subject-balance cap (maximum 96 windows per subject)	2,714	186,252	77.5%

Table S3: Retained-versus-excluded distributions for the primary PulseDB-Vital cohort.

Window group	Windows	SBP median [IQR]	DBP median [IQR]	Age median [IQR]	Female proportion	Min-modality SQI; apparent PAT median [IQR]
Final retained after cap	186,252	126 [117–135]	68 [63–75]	61 [51–69]	36.3%	0.69 [0.60–0.78]; 277 [233–320] ms
Removed by fixed eligibility checks	32,003	125 [116–135]	68 [62–74]	61 [51–69]	34.9%	0.40 [0.33–0.48]; 278 [234–321] ms
Removed only by 96-window cap	7,281	126 [117–135]	68 [63–74]	61 [51–69]	34.1%	0.73 [0.64–0.81]; 277 [234–320] ms

Table S4: Reproducibility summary for the subject-disjoint benchmark.

Item	Setting	Supporting repository record
Outer folds	Five fixed subject-grouped folds; 542–543 test subjects, 271 validation subjects, and 1,900–1,901 training subjects per fold; every retained subject is tested exactly once	Fold manifests
Partition integrity	Train, validation, and test subjects are disjoint in every fold; each retained subject is tested exactly once	Partition and membership audits
Validation assignment	In each fold, 271 fixed-seed subjects from the four non-test groups form the validation set; all remaining non-test subjects train the model	Fold-membership policy and validation manifest
Training scope	All main models use the same deterministic whole-window robust normalization; structural parameters are fixed before outer evaluation; validation subjects were used for the common training-setting grid, early stopping, and checkpoint selection	Design-provenance and fold-local selection records
Prediction files	7 released prediction artifacts support the main, mechanism, and transfer analyses	Prediction manifest
Released checkpoints	10 representative outer-fold checkpoints and 3 frozen source-model checkpoints are provided for executable review	Checkpoint manifest
Executable code	Raw-source adapters, fold construction, training, inference, metric recomputation, and figure reproduction are released	Repository reproduction entry point
Integrity	Package files are covered by a deterministic SHA-256 inventory and the public workflow is rerun in a clean output directory	Hash inventory and clean-room report

Table S5: PhysioCAT architecture and training configuration used in the primary subject-disjoint benchmark. Structural choices were fixed before outer evaluation; conventional training settings were selected within each outer fold from a fixed three-candidate grid using validation subjects only.

Component	Frozen setting
Input and tokenization	Paired 8 s ECG/PPG windows at 250 Hz; 2,000 samples; non-overlapping 16-sample patches; 125 tokens; 64 ms token spacing
Model-input normalization	All main models and comparators: whole-window 0.5th–99.5th percentile pre-clipping followed by median/IQR scaling with floor 10^{-6} and clipping to $[-8, 8]$; no subject- or cohort-fitted statistics. Patch-local normalization is used only as a sensitivity control
Analysis branch	Zero-phase fourth-order Butterworth filters (ECG 0.5–40 Hz; PPG 0.4–12 Hz) followed by level-4 Symlet-4 soft-threshold denoising for beat detection and SQI only; the model branch uses the same shared crop without this analysis transform, followed by its declared deterministic normalization
Patch encoders	Modality-specific 128-channel local projection; token-wise residual MLP expansion $\times 2$; learned position and modality embeddings
Cross-attention	Four heads; opposite query/key assignments score the same ECG-leading edges; support-conservative offsets 3–6 (center lags 192–384 ms; complete patch support 132–444 ms inside the 120–450 ms literature envelope); 482 edges and 122 active ECG-anchor rows; reciprocal affinities and pairwise ECG/PPG SQI are fused once on each admissible edge; the three unsupported boundary rows are exactly zero
Temporal context and head	Two post-fusion Transformer layers; unsupported boundary rows are padding-masked and excluded from attentive/mean/max pooling; feed-forward expansion $\times 4$; dropout 0.10; 128-unit GELU regression layer; two outputs
Optimization	AdamW; selected learning rate 6.0e-04, weight decay 0.0e+00, and batch size 32 from a fixed three-candidate grid using validation mean SBP/DBP MAE in each outer fold
Schedule and checkpointing	Five-epoch linear warm-up followed by cosine decay; at most 80 supervised epochs; patience 12; minimum validation mean SBP/DBP MAE, earliest epoch breaking ties
Contrastive pre-training	5 fold-local epochs; mean-pooled 128-dimensional ECG and PPG features; separate training-only two-layer 128-dimensional projection heads; intra-ECG, intra-PPG, and cross-modal InfoNCE weights 1:1:0.5; temperature 0.10; shared paired shift ± 8 samples; independent amplitude perturbation SD 0.04 and additive-noise SD 0.025; same-subject off-diagonal negatives excluded
Supervised objective	Component-wise Huber $\delta = 5$ mmHg; soft SBP–DBP non-inversion penalty with weight 0.15; MuFuBP-Net alone retains its published probabilistic KL term at weight 10^{-3}
Seeds and environment	Initialization/data-order seed 42; random-sparse control seeds 82,000–82,019; deterministic-algorithm mode; PyTorch 2.12.0; profiling environments are listed with the released timing logs

Supervised loss. For prediction $\hat{y} = (\hat{s}, \hat{d})$ and target y, the optimized scalar-target objective was

$$L_{sup} = L_{Huber,\delta=5}(\hat{y}, y) + 0.15[\hat{d} - \hat{s}]_{+}.$$

The contrastive stage applied separate training-only two-layer projection heads to the mean-pooled ECG and PPG features. The intra-ECG, intra-PPG, and synchronized ECG–PPG InfoNCE terms were weighted 1:1:0.5, and same-subject off-diagonal pairs were excluded from every denominator. Projection heads were discarded before supervised inference.

Table S6: Comparator implementation and checkpoint summary for the main comparison. Parameter counts are computed directly from the released model classes. All methods use the same whole-window robust normalization and scalar-output protocol; the matched control shares the complete PhysioCAT parameterization and replaces only the delay-band restriction with full-key attention on the same query support.

Method	Role	Input adaptation	Params.	Implementation / adaptation	Checkpoint selection	SBP MAE	DBP MAE
PhysioCAT	Proposed model	ECG + PPG + token SQI; window-robust	0.77M	Local patch encoders; reciprocal query/key scores combined on the same ECG-leading edges; pairwise ECG/PPG SQI; post-fusion temporal context	Fold-local validation for training-setting and checkpoint selection	4.26	3.11
No-delay-band cross-attention	Matched mechanism control	ECG + PPG + token SQI; window-robust	0.77M	Identical reciprocal edge-aligned backbone and training configuration; unrestricted key support on the default ECG-anchor rows	Fold-local validation for training-setting and checkpoint selection	5.25	3.51
MuFuBP-Net	Neural comparator	ECG + PPG; window-robust	7.06M	Morphology/dynamics streams and probabilistic fusion retained; scalar head and subject-disjoint folds adapted	Fold-local validation for checkpoint selection	5.10	3.38
TE-SAGRU	Neural comparator	ECG + PPG; window-robust	2.17M	Parallel modality encoders and stacked attention-gated GRU fusion retained; raw-waveform front end and scalar head adapted	Fold-local validation for checkpoint selection	5.53	3.70
BP-Net	Neural comparator	ECG + PPG; window-robust	1.94M	Deep convolutional BP feature extractor retained; paired-input and two-output head adapted	Fold-local validation for checkpoint selection	6.08	4.17
CNN-BiLSTM	Neural comparator	ECG + PPG; window-robust	1.26M	Multiscale CNN, bidirectional LSTM, and common two-output head	Fold-local validation for checkpoint selection	6.37	4.34
Engineered-feature random forest	Feature comparator	ECG + PPG features; window-robust	500 trees	500-tree fold-local estimator with deterministic engineered features	Fold-local fitting	8.32	5.20
PAT ridge model	Feature comparator	PAT + heart rate + missing-PAT indicator; window-robust	8 coeff.	Three features plus one bias per outcome; eight fitted coefficients in total	Fold-local fitting	9.52	6.06

Table S7: Optimization- and mask-topology stability on the complete five-fold subject-grouped PulseDB-Vital protocol. The first four rows vary training seeds 42, 1337, and 2025; the final row summarizes 20 independently trained degree-preserving random-rewiring topologies with fixed initialization and data-order seeds. Values are window-weighted MAE mean ± standard deviation across seeds.

Stability condition	SBP MAE (mmHg)	DBP MAE (mmHg)
PhysioCAT (3 training seeds)	4.25 ± 0.04	3.13 ± 0.02
MuFuBP-Net (3 training seeds)	5.14 ± 0.01	3.39 ± 0.01
No-delay-band cross-attention (3 training seeds)	5.28 ± 0.04	3.55 ± 0.01
PPG-leading mirror delay band (3 training seeds)	4.86 ± 0.03	3.47 ± 0.03
Degree-preserving random rewirings (20 independently trained topologies)	5.64 ± 0.05	3.62 ± 0.04

S2. Protocol robustness and threshold checks

Table S8: Protocol-sensitivity control on PulseDB-Vital: five-fold out-of-fold random segment-level evaluation versus five-fold subject-grouped out-of-fold testing.

Evaluation protocol	Train–test subject overlap	PhysioCAT SBP MAE	PhysioCAT DBP MAE	No-delay-band control SBP MAE	No-delay-band control DBP MAE
Five-fold OOF random segment-level split	$> 99\%$	2.23	1.74	3.22	2.02
Five-fold subject-grouped OOF testing	0%	4.26	3.11	5.25	3.51

Table S9: Subject-level errors on the synchronized PulseDB-Vital subject-disjoint evaluation.

Subject-level metric	SBP	DBP
Mean MAE, mmHg	4.27	3.12
Median MAE (IQR), mmHg	4.19 (3.70–4.75)	3.07 (2.74–3.45)
90th percentile MAE, mmHg	5.34	3.83
Subjects with MAE ≤ 5 mmHg	82.5%	99.7%

Table S10: Subject-cluster bootstrap 95% confidence intervals for MAE. Window estimates retain all windows from each resampled held-out subject; subject-weighted estimates first average within subject and then across subjects. All three core models use 2,000 resamples with seed 42.

Cohort	Model	Window SBP MAE [95% cluster CI]	Window DBP MAE [95% cluster CI]	Subject-weighted SBP MAE [95% CI]	Subject-weighted DBP MAE [95% CI]
PulseDB-Vital	PhysioCAT	4.26 [4.23, 4.29]	3.11 [3.09, 3.13]	4.27 [4.24, 4.30]	3.12 [3.10, 3.14]
PulseDB-Vital	Matched no-delay-band cross-attention	5.25 [5.20, 5.30]	3.51 [3.48, 3.54]	5.28 [5.23, 5.34]	3.53 [3.50, 3.57]
PulseDB-Vital	MuFuBP-Net	5.10 [5.05, 5.16]	3.38 [3.35, 3.41]	5.14 [5.09, 5.20]	3.40 [3.37, 3.43]
PulseDB-MIMIC	PhysioCAT	4.95 [4.91, 5.00]	3.52 [3.49, 3.54]	4.98 [4.94, 5.02]	3.54 [3.51, 3.57]
PulseDB-MIMIC	Matched no-delay-band cross-attention	6.40 [6.33, 6.48]	4.20 [4.16, 4.25]	6.46 [6.38, 6.53]	4.24 [4.20, 4.29]
PulseDB-MIMIC	MuFuBP-Net	6.15 [6.08, 6.22]	4.12 [4.07, 4.16]	6.21 [6.14, 6.28]	4.14 [4.09, 4.18]
MIMIC-BP	PhysioCAT	4.83 [4.77, 4.89]	3.41 [3.37, 3.45]	4.83 [4.77, 4.90]	3.42 [3.38, 3.46]
MIMIC-BP	Matched no-delay-band cross-attention	6.45 [6.36, 6.54]	4.22 [4.17, 4.28]	6.45 [6.35, 6.54]	4.23 [4.18, 4.29]
MIMIC-BP	MuFuBP-Net	5.96 [5.88, 6.04]	4.03 [3.98, 4.09]	5.98 [5.89, 6.06]	4.03 [3.98, 4.09]

The no-delay-band PhysioCAT control is the main mechanism comparator for the delay constraint. Baseline modules, configurations, primary prediction lineage, metric source tables, and local verification manifests are documented in the public reproducibility repository.

Table S11: Error-threshold analysis on the synchronized PulseDB-Vital cohort.

Metric	SBP	DBP
ME (mmHg)	0.17	0.16
MAE (mmHg)	4.26	3.11
SDE (mmHg)	5.48	3.98
RMSE (mmHg)	5.48	3.98
Absolute error $\leq$ 5 mmHg	66.3%	80.4%
Absolute error $\leq$ 10 mmHg	93.0%	98.3%
Absolute error $\leq$ 15 mmHg	98.9%	99.9%
Bland–Altman 95% LoA (mmHg)	-10.6 to 10.9	-7.6 to 8.0

Table S12: Internal and external error profiles under the fixed retrospective protocols.

A. PhysioCAT error-threshold profiles

Cohort	SBP ME $\pm$ SDE	DBP ME $\pm$ SDE	Absolute error $\leq$5, 10, 15 mmHg (SBP)	Absolute error $\leq$5, 10, 15 mmHg (DBP)
PulseDB-Vital	0.17 ± 5.48	0.16 ± 3.98	66.3, 93.0, 98.9%	80.4, 98.3, 99.9%
PulseDB-MIMIC	0.09 ± 6.41	-0.02 ± 4.52	59.8, 88.8, 97.4%	75.1, 96.8, 99.7%
MIMIC-BP	0.12 ± 6.19	0.01 ± 4.35	60.6, 89.8, 97.9%	76.1, 97.5, 99.8%

B. External matched-control and neural-comparator profiles

Cohort	Comparator	SBP ME $\pm$ SDE	DBP ME $\pm$ SDE	SBP MAE (r)	DBP MAE (r)
PulseDB-MIMIC	Matched no-delay-band cross-attention	0.22 ± 8.36	0.12 ± 5.45	6.40 (0.84)	4.20 (0.85)
PulseDB-MIMIC	MuFuBP-Net	-0.03 ± 8.07	0.15 ± 5.38	6.15 (0.86)	4.12 (0.85)
MIMIC-BP	Matched no-delay-band cross-attention	0.15 ± 8.41	0.05 ± 5.48	6.45 (0.85)	4.22 (0.84)
MIMIC-BP	MuFuBP-Net	-0.08 ± 7.70	0.13 ± 5.22	5.96 (0.87)	4.03 (0.86)

S3. Extended model analyses

Table S13: Additional input, training, fusion, and normalization controls under the complete five-fold subject-grouped protocol.

A. Input, training, and fusion controls

Control family	Variant	SBP MAE	DBP MAE
Input	ECG only	8.54	5.74
Input	PPG only	7.09	4.49
Training	Without subject-aware pre-training	4.68	3.31
Fusion	Early concatenation	5.98	3.85
Fusion	Late averaging	6.19	3.99
Fusion	Gated fusion	5.89	3.82
Fusion	SE-based fusion	5.88	3.78
Fusion	No-delay-band bidirectional attention	5.25	3.51
Fusion	PPG-leading mirror delay band	4.82	3.45
Fusion	Delay-banded unidirectional attention	4.87	3.49
Fusion	Full PhysioCAT	4.26	3.11

B. Matched 2×2 normalization factorial

Configuration	SBP MAE	DBP MAE
Delay band + whole-window robust normalization	4.26	3.11
No-delay + whole-window robust normalization	5.25	3.51
Delay band + patch-local normalization	4.45	3.23
No-delay + patch-local normalization	5.40	3.59

Table S14: Inference-time negative-control experiments on the synchronized PulseDB-Vital cohort. Increases are calculated separately for each model relative to its own unperturbed synchronized baseline.

Perturbation	No-delay SBP/DBP MAE	PhysioCAT SBP/DBP MAE	PhysioCAT increase SBP/DBP	Interpretation
None	5.25 / 3.51	4.26 / 3.11	–	Normal synchronized input
Token-aligned circular PPG shift (3–10 complete 16-sample tokens; 192–640 ms)	7.04 / 4.53	8.32 / 4.97	+4.06 / +1.86	Moves complete PPG token blocks while preserving within-token samples
Cross-subject ECG-PPG pairing	7.37 / 4.69	8.78 / 5.24	+4.52 / +2.13	Breaks subject-specific coupling
PPG time reversal	7.21 / 4.55	8.47 / 5.08	+4.21 / +1.97	Destroys pulse morphology order
Implausible delay mask	6.80 / 4.34	7.66 / 4.71	+3.39 / +1.60	Forces non-physiological attention

Table S15: Performance across signal-quality strata on the synchronized primary cohort.

SQI stratum	Windows (share)	PhysioCAT SBP MAE	PhysioCAT DBP MAE	Source record
High SQI (minimum SQI ≥ 0.85)	21,329 (11.5%)	4.13	2.84	Released high-SQI prediction slice
Medium SQI ($0.70 \leq$ minimum SQI < 0.85)	67,952 (36.5%)	4.19	2.97	Released medium-SQI prediction slice
Low SQI ($0.40 \leq$ minimum SQI < 0.70; maximum modality SQI ≥ 0.55)	96,971 (52.1%)	4.34	3.26	Released low-SQI prediction slice under the joint retention rule

Table S16: Delay-band sensitivity under the complete five-fold subject-grouped protocol. The literature-guided 120–450 ms band was held fixed throughout model comparison.

Delay band	SBP MAE	DBP MAE	Interpretation
120–300 ms	4.55	3.28	Narrow support-conservative band
120–350 ms	4.43	3.19	Strong but slightly restrictive
120–450 ms	4.26	3.11	Literature-guided fixed band
80–550 ms	4.45	3.22	Less restrictive
No-delay-band cross-attention	5.25	3.51	No physiological timing prior

Table S17: Equal-width delay-location sweep on a common query support. Every band contains four offsets, 113 ECG-anchor rows, and 452 admissible edges; only delay location changes.

Offset band	ECG-anchor rows	Edges	SBP MAE	DBP MAE
offsets 2–5	113	452	4.51	3.26
offsets 3–6	113	452	4.33	3.18
offsets 4–7	113	452	4.50	3.25

S4. Transfer, subgroup, quality, and efficiency analyses

Residual-offset calibration estimated a shrunken mean reference-minus-prediction residual for each eligible MIMIC-BP subject from 1, 2, 5, or 10 calibration windows, with a five-window prior strength. Results summarize 200 fixed-seed calibration-window draws on a common evaluation set; the reported ranges describe those draws rather than population uncertainty.

Input-quality sensitivity applied the same fitted outer-fold model to nested target-formed window sets while varying the ECG/PPG SQI rule or subject-balance cap. The easier and harder waveform examples were selected by fixed descriptive rules from the high-SQI regular-rhythm and low-SQI RR-irregular pools, respectively.

The source-shift map used two-component PCA of standardized demographic, SQI, spectral, and pulse-shape summaries. Runtime profiling used batch-1 inference after 500 warm-up windows and included signal preparation and tokenization in the end-to-end measurements.

Table S18: Few-shot subject residual-offset calibration on MIMIC-BP after zero-shot transfer from PulseDB-Vital. Results cover 1,522 eligible subjects and 200 fixed-seed calibration-window draws; brackets are the 2.5th–97.5th percentiles across draws, not population confidence intervals, and calibration windows were excluded from the common evaluation set.

Calibration windows	Offset estimator	Evaluation windows	PhysioCAT SBP MAE	PhysioCAT DBP MAE	MuFuBP-Net SBP MAE	MuFuBP-Net DBP MAE
0	none	24,264	4.82 [4.80, 4.85]	3.41 [3.38, 3.43]	5.95 [5.92, 5.98]	4.04 [4.01, 4.06]
1	shrunken subject residual mean (prior strength 5 windows)	24,264	4.76 [4.72, 4.79]	3.37 [3.35, 3.40]	5.86 [5.82, 5.89]	3.98 [3.95, 4.01]
2	shrunken subject residual mean (prior strength 5 windows)	24,264	4.71 [4.68, 4.75]	3.35 [3.32, 3.37]	5.79 [5.76, 5.83]	3.94 [3.91, 3.97]
5	shrunken subject residual mean (prior strength 5 windows)	24,264	4.62 [4.59, 4.65]	3.30 [3.28, 3.32]	5.67 [5.63, 5.71]	3.86 [3.83, 3.89]
10	shrunken subject residual mean (prior strength 5 windows)	24,264	4.55 [4.52, 4.58]	3.25 [3.23, 3.28]	5.58 [5.55, 5.61]	3.80 [3.78, 3.83]

Table S19: Subgroup definitions and performance values used in Supplementary Fig. S2. BP and quality strata were defined at the window level; a subject could contribute to more than one stratum.

Family	Subgroup	Definition	Subjects	Windows	Proportion	SBP MAE	DBP MAE
Age	18–40	$18 \leq$ age ≤ 40 years	182	12,123	6.5%	4.26	3.18
Age	41–65	$41 \leq$ age ≤ 65 years	1,554	107,135	57.5%	4.24	3.11
Age	66+	Age ≥ 66 years	978	66,994	36.0%	4.30	3.09
Sex	Female	Sex recorded as female	985	67,565	36.3%	4.21	3.05
Sex	Male	Sex recorded as male	1,729	118,687	63.7%	4.29	3.14
BP	Lower-BP windows	SBP < 130 mmHg and DBP < 80 mmHg; window stratum, not a clinical diagnosis	2,710	111,741	60.0%	4.24	3.13
BP	Elevated-BP windows	SBP ≥ 130 mmHg or DBP ≥ 80 mmHg; window stratum, not a clinical diagnosis	2,703	74,511	40.0%	4.30	3.08
Rhythm	Normal rhythm	$\sqrt{CV(RR)^2 + [RMSSD/\mathrm{median}(RR)]^2} < 0.28$	2,696	180,213	96.8%	4.25	3.10
Rhythm	Irregular rhythm	$\sqrt{CV(RR)^2 + [RMSSD/\mathrm{median}(RR)]^2} \geq 0.28$; descriptive rhythm-irregularity label, not a clinical arrhythmia diagnosis	274	6,039	3.2%	4.52	3.20
SQI	Lowest SQI	$0.40 \leq$ minimum SQI < 0.70, with maximum modality SQI ≥ 0.55	2,714	96,971	52.1%	4.34	3.26

Table S20: Primary repeated-measures Bland–Altman agreement for PhysioCAT on PulseDB-Vital. Limits combine within- and between-subject residual variance; residual ICC reports the between-subject share of total residual variance.

Dataset	Windows	SBP ME	SBP repeated-measures 95% LoA	SBP residual ICC	DBP ME	DBP repeated-measures 95% LoA	DBP residual ICC
PulseDB-Vital	186,252	0.17	-10.6 to 10.9	0.050	0.16	-7.6 to 8.0	0.043

Supplementary Table S21 reports the matched model footprint and measured batch-1 processing profile for representative server, workstation, and compact-edge settings.

Table S21: Model footprint and measured post-acquisition processing profile. Batch-1 runtimes are medians of 5,000 samples after 500 warm-up windows; the 8 s acquisition interval is excluded. On the fixed INT8 agreement set, SBP/DBP MAE changed by 0.04/0.02 mmHg relative to FP16.

A. Matched model footprint on H800

Model	Parameters	FLOPs / 8 s window	Forward latency (ms)	FP32 parameter memory
No-delay-band cross-attention	0.77M	0.38G	1.84	3.1 MB
PhysioCAT	0.77M	0.38G	1.86	3.1 MB

B. Cross-platform PhysioCAT processing profile

Platform	Runtime	Forward latency (ms)	End-to-end latency (ms)	Throughput (windows/s)	8 s window / processing time
NVIDIA H800 80 GB	PyTorch FP16	1.86	5.94	168.4	×1347
Intel Xeon Gold 6338	ONNX Runtime FP32	31.70	74.60	13.4	×107
Jetson Orin NX 16 GB	TensorRT FP16	8.94	19.80	50.5	×404
Jetson Orin NX 16 GB	TensorRT INT8	5.21	12.60	79.4	×635

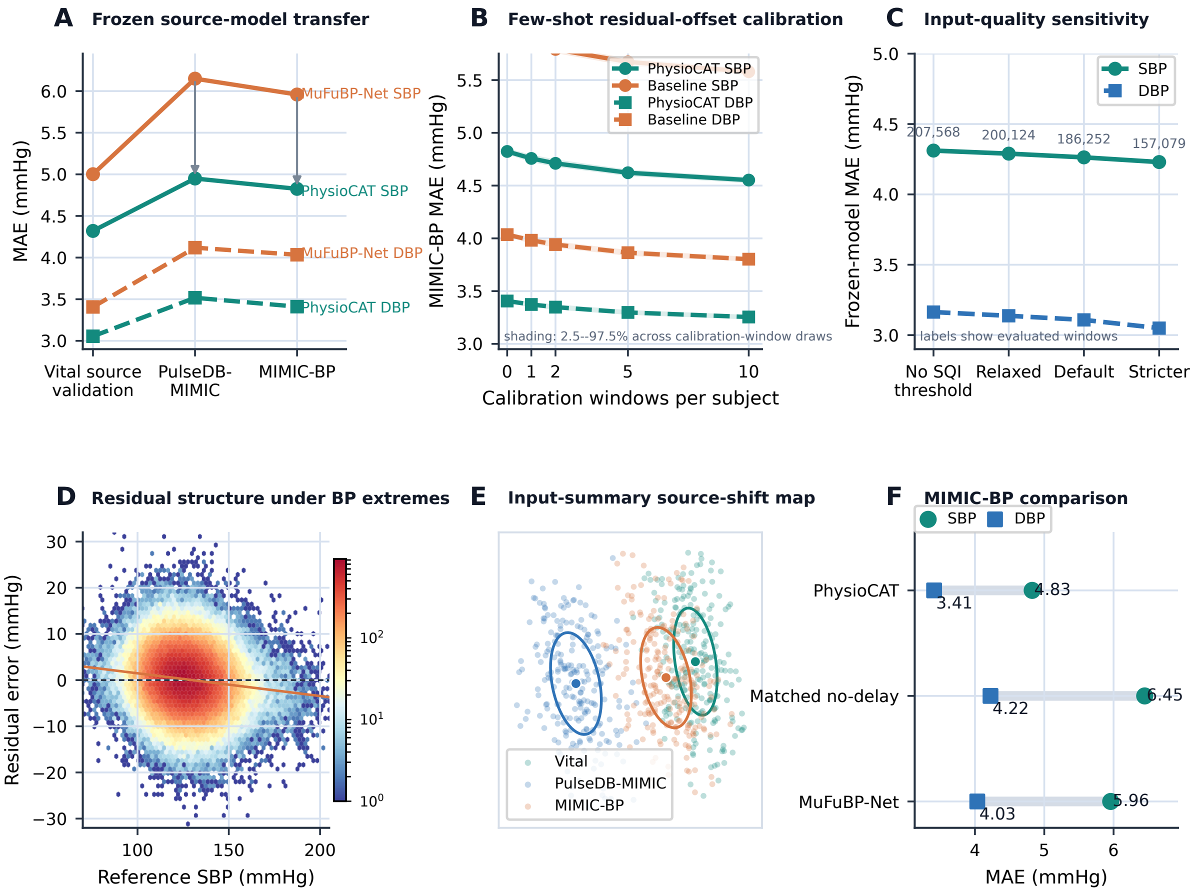

Figure S1: External-transfer and secondary analyses: frozen source-model validation and zero-shot transfer, residual-offset calibration, input-quality inclusion sensitivity, SBP residual structure, released input-summary source shift, and MIMIC-BP comparison.

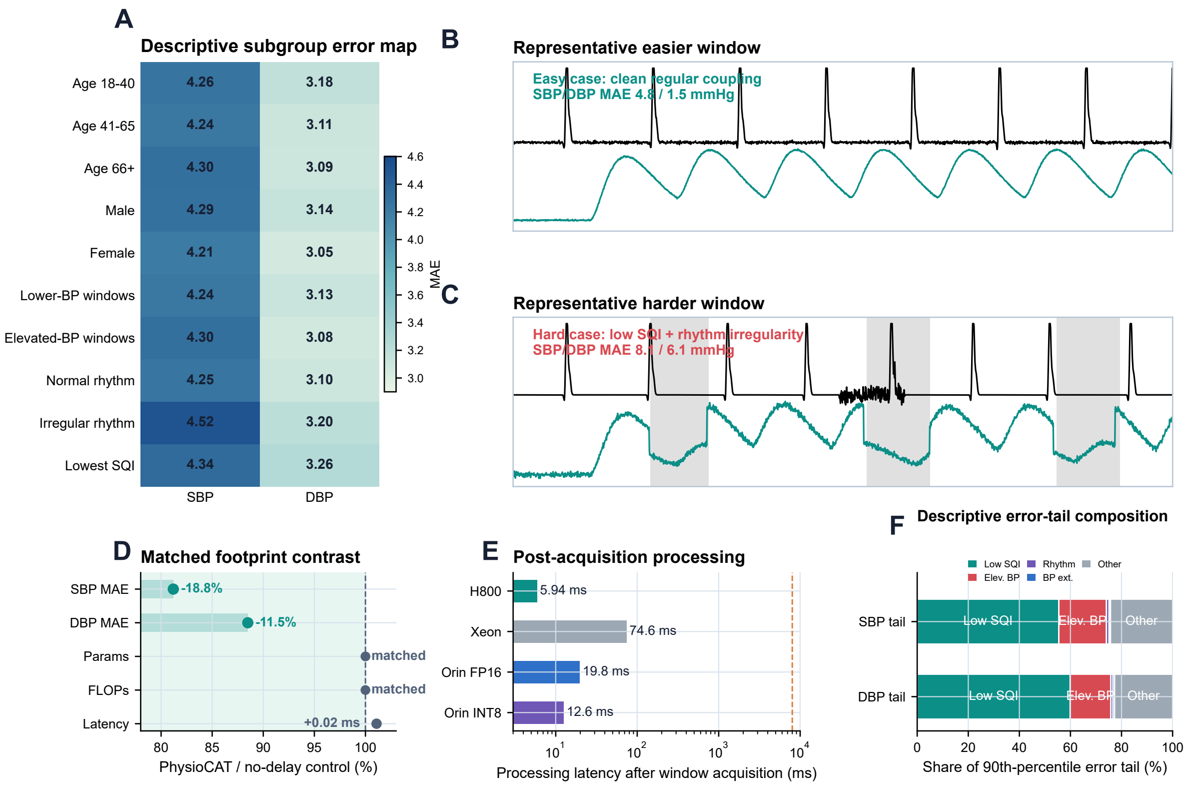

Figure S2: Error-structure and computational-profile analyses: subgroup errors, released representative easier and harder waveform windows, matched-footprint comparison, post-acquisition processing latency, and descriptive high-error-tail composition.

S5. Additional mechanism and validation analyses

Table S22: Sparse and structured mask controls under the complete five-fold subject-grouped protocol. The degree-preserving random-rewiring row uses the fixed topology seed 82,000; the stability analysis separately summarizes 20 independently trained topologies.

Mask configuration	Allowed-edge density	ECG-leading timing	SBP MAE	DBP MAE	Window-weighted Δ MAE vs full (SBP/DBP)	Subject-paired Holm p (SBP/DBP)	Interpretation
No-delay-band cross-attention	Not sparse	No	5.25	3.51	+0.99 / +0.40	$<$ 0.001 / $<$ 0.001	Search-space control without a delay prior
Fixed-uniform aggregation within the default delay band	Same admissible edges as the default band	Yes	4.51	3.27	+0.25 / +0.16	$<$ 0.001 / $<$ 0.001	The physiological band alone is insufficient without learned within-band edge weighting
Degree-preserving random rewiring	Exactly preserves the default row and column degree vectors	No	5.63	3.65	+1.37 / +0.54	$<$ 0.001 / $<$ 0.001	Sparse graph capacity alone does not recover delay-aware performance
Same-width shifted band	452 edges on the same 113 ECG anchors used by the equal-width location sweep	Wrong delay region	5.52	3.65	+1.25 / +0.55	$<$ 0.001 / $<$ 0.001	Moving an otherwise equal-capacity band away from plausible pulse-arrival timing is suboptimal
Zero-centered nonzero local mask	494 natural-boundary edges; zero lag excluded	No	5.40	3.54	+1.14 / +0.43	$<$ 0.001 / $<$ 0.001	Generic local cross-modal proximity is insufficient
PPG-leading mirror delay band	Same parameters and total edge count as the default band	No; encodes the opposite ordering	4.82	3.45	+0.56 / +0.34	$<$ 0.001 / $<$ 0.001	Reversing the ECG-leading relation weakens the matched backbone
Default delay-banded asymmetric attention (PhysioCAT)	Natural boundary-aware support	Yes	4.26	3.11	– / –	– / –	Default PhysioCAT mask

Table S23: Complete held-out-subject paired comparison family. Positive reductions are comparator MAE minus PhysioCAT MAE; confidence intervals are subject-cluster bootstrap intervals and p values are two-sided Wilcoxon tests with Holm adjustment across the fixed 22-test family.

Cohort	Comparator	Outcome	MAE reduction (mmHg)	95% CI	Holm-adjusted p	Rank-biserial effect
PulseDB-Vital	No-delay-band cross-attention	SBP	1.01	[0.96, 1.07]	< 0.001	0.67
PulseDB-Vital	No-delay-band cross-attention	DBP	0.42	[0.38, 0.45]	< 0.001	0.47
PulseDB-Vital	MuFuBP-Net	SBP	0.87	[0.81, 0.93]	< 0.001	0.56
PulseDB-Vital	MuFuBP-Net	DBP	0.28	[0.25, 0.32]	< 0.001	0.30
PulseDB-Vital	Fixed-uniform aggregation within the default delay band	SBP	0.26	[0.22, 0.30]	< 0.001	0.28
PulseDB-Vital	Fixed-uniform aggregation within the default delay band	DBP	0.16	[0.14, 0.19]	< 0.001	0.26
PulseDB-Vital	PPG-leading mirror delay band	SBP	0.57	[0.53, 0.62]	< 0.001	0.49
PulseDB-Vital	PPG-leading mirror delay band	DBP	0.34	[0.32, 0.37]	< 0.001	0.50
PulseDB-Vital	Zero-centered nonzero local mask	SBP	1.16	[1.10, 1.22]	< 0.001	0.74
PulseDB-Vital	Zero-centered nonzero local mask	DBP	0.44	[0.40, 0.47]	< 0.001	0.49
PulseDB-Vital	Shifted offsets 9–12 (516–828 ms support)	SBP	1.28	[1.23, 1.35]	< 0.001	0.78
PulseDB-Vital	Shifted offsets 9–12 (516–828 ms support)	DBP	0.55	[0.51, 0.59]	< 0.001	0.58
PulseDB-Vital	Degree-preserving random rewiring	SBP	1.39	[1.32, 1.44]	< 0.001	0.80
PulseDB-Vital	Degree-preserving random rewiring	DBP	0.54	[0.51, 0.58]	< 0.001	0.58
PulseDB-MIMIC	No-delay-band cross-attention	SBP	1.48	[1.40, 1.56]	< 0.001	0.75
PulseDB-MIMIC	No-delay-band cross-attention	DBP	0.70	[0.65, 0.75]	< 0.001	0.62
PulseDB-MIMIC	MuFuBP-Net	SBP	1.23	[1.16, 1.31]	< 0.001	0.65
PulseDB-MIMIC	MuFuBP-Net	DBP	0.60	[0.54, 0.65]	< 0.001	0.53
MIMIC-BP	No-delay-band cross-attention	SBP	1.62	[1.52, 1.72]	< 0.001	0.76
MIMIC-BP	No-delay-band cross-attention	DBP	0.82	[0.75, 0.88]	< 0.001	0.66
MIMIC-BP	MuFuBP-Net	SBP	1.15	[1.04, 1.25]	< 0.001	0.58
MIMIC-BP	MuFuBP-Net	DBP	0.62	[0.54, 0.68]	< 0.001	0.50

Table S24: Input-quality validity, inclusion sensitivity, and quality-aware deferral on PulseDB-Vital. All performance rows use the frozen subject-grouped outer models.

A. Retention-rule and subject-cap sensitivity

Evaluation view	Subjects	Windows	SBP MAE	DBP MAE
Basic paired/target validity; no SQI threshold	2,714	207,568	4.31	3.16
Relaxed joint SQI (min 0.30; max 0.45)	2,714	200,124	4.29	3.14
Default joint SQI (min 0.40; max 0.55)	2,714	186,252	4.26	3.11
Stricter joint SQI (min 0.50; max 0.65)	2,682	157,079	4.23	3.05
Default joint SQI; 48-window cap	2,714	124,929	4.26	3.11
Default joint SQI; no subject cap	2,714	193,533	4.26	3.11

B. Quality-aware deferral

Deferred fraction	Remaining coverage	Remaining windows	SBP MAE	DBP MAE
0%	100%	186,252	4.26	3.11
5%	95%	176,939	4.25	3.08
10%	90%	167,627	4.24	3.07
15%	85%	158,314	4.23	3.05
20%	80%	149,002	4.22	3.03

C. Blinded paired-waveform quality review

Windows	Subjects	Usable	Rater agreement	Cohen κ	Min-SQI AUC	Rule sensitivity / specificity
1,200	968	89.2%	0.932	0.660	0.986	0.947 / 0.862

Table S25: Performance across fixed reference-BP strata on PulseDB-Vital. Strata were defined before analysis and retain the same out-of-fold PhysioCAT predictions; values are window-weighted.

Outcome	Reference stratum	Subjects	Windows	Reference mean	ME	MAE	SDE
SBP	< 90	726	963	83.2	2.29	4.74	5.67
SBP	90–109	2,377	20,100	104.3	1.26	4.42	5.53
SBP	110–129	2,709	92,927	121.0	0.43	4.20	5.38
SBP	130–149	2,693	62,871	137.7	-0.38	4.26	5.47
SBP	150–179	2,137	8,834	157.5	-1.32	4.49	5.63
SBP	≥ 180	478	557	189.4	-2.97	5.36	6.20
DBP	< 50	1,139	1,666	47.0	1.38	3.41	4.07
DBP	50–69	2,711	104,763	63.1	0.46	3.13	3.98
DBP	70–89	2,709	76,282	76.3	-0.21	3.05	3.90
DBP	90–109	1,662	3,133	95.6	-1.12	3.36	4.15
DBP	≥ 110	373	408	118.5	-2.00	3.76	4.39

Table S26: Apparent-PAT interaction across the fixed timing prior. Apparent PAT is a post-retention descriptive variable and does not select windows or alter inference. Parentheses show comparator MAE minus PhysioCAT MAE; positive values favor PhysioCAT.

Apparent-PAT group	Model	Windows	SBP MAE (Δ)	DBP MAE (Δ)
Detected, 120–450 ms	PhysioCAT	173,379	4.21 (+0.00)	3.07 (+0.00)
Detected, 120–450 ms	Matched no-delay	173,379	5.23 (+1.01)	3.50 (+0.43)
Detected, 120–450 ms	PPG-leading mirror band	173,379	4.78 (+0.57)	3.43 (+0.36)
Detected, 120–450 ms	Zero-centered nonzero local	173,379	5.39 (+1.17)	3.53 (+0.46)
Detected, outside 120–450 ms	PhysioCAT	4,864	4.45 (+0.00)	3.32 (+0.00)
Detected, outside 120–450 ms	Matched no-delay	4,864	5.24 (+0.78)	3.58 (+0.26)
Detected, outside 120–450 ms	PPG-leading mirror band	4,864	4.94 (+0.49)	3.55 (+0.23)
Detected, outside 120–450 ms	Zero-centered nonzero local	4,864	5.42 (+0.97)	3.56 (+0.24)
Not detected	PhysioCAT	8,009	5.22 (+0.00)	3.82 (+0.00)
Not detected	Matched no-delay	8,009	5.77 (+0.55)	3.81 (-0.01)
Not detected	PPG-leading mirror band	8,009	5.63 (+0.41)	3.91 (+0.09)
Not detected	Zero-centered nonzero local	8,009	5.77 (+0.55)	3.79 (-0.03)

Table S27: Source-model and external-protocol validation before and after zero-shot transfer.

A. Frozen PulseDB-Vital source split and internal validation

Model	Train subjects	Train windows	Validation subjects	Validation windows	SBP MAE	DBP MAE
PhysioCAT	2,442	167,483	272	18,769	4.32	3.06
Matched no-delay	2,442	167,483	272	18,769	5.41	3.53
MuFuBP-Net	2,442	167,483	272	18,769	5.00	3.41

B. Zero-shot source-training-seed stability

Target protocol	Source model	Seeds	SBP MAE	DBP MAE
PulseDB-MIMIC	PhysioCAT	3	4.96 ± 0.04	3.54 ± 0.02
PulseDB-MIMIC	Matched no-delay	3	6.41 ± 0.03	4.19 ± 0.03
PulseDB-MIMIC	MuFuBP-Net	3	6.15 ± 0.02	4.12 ± 0.02
MIMIC-BP	PhysioCAT	3	4.81 ± 0.03	3.41 ± 0.01
MIMIC-BP	Matched no-delay	3	6.43 ± 0.07	4.22 ± 0.01
MIMIC-BP	MuFuBP-Net	3	6.00 ± 0.04	4.05 ± 0.03

C. Identity separation of the two MIMIC-derived target protocols

Target protocols	Cohort A subjects	Cohort B subjects	Shared patient hashes	Shared record hashes
PulseDB-MIMIC / MIMIC-BP	2,201	1,524	0	0

Table S28: Target-reference construction and sensitivity on PulseDB-Vital. The blinded review in panel E used a fixed-seed, threshold-enriched sample spanning four ABP-SQI strata; its usable percentage describes the review sample rather than cohort prevalence.

A. Concordance of synchronized ABP-beat targets with repository scalar references

Scope	Outcome	Windows	Alternative reference	Bias	MAE	SD of difference
target-formed candidate pool	SBP	225,536	repository_scalar	-0.01	0.38	0.48
target-formed candidate pool	DBP	225,536	repository_scalar	0.01	0.29	0.36
final retained cohort	SBP	186,252	repository_scalar	-0.01	0.38	0.48
final retained cohort	DBP	186,252	repository_scalar	0.01	0.29	0.36

B. Target-formation comparison

Group	Paired candidates	Repository label available	Repository SBP median [IQR]	Repository DBP median [IQR]	Age median [IQR]	Female
Target formed	225,536	225,536 (100.0%)	125.9 [116.9–135.4]	68.3 [62.7–74.4]	61 [51–69]	36.0%
Target-formation failure	9,531	8,451 (88.7%)	126.3 [117.2–135.5]	68.4 [62.9–74.5]	61 [52–70]	36.6%

Group	Min-SQI median [IQR]	RR irregularity median [IQR]	Max |SMD|
Target formed	0.67 [0.55–0.77]	0.13 [0.09–0.18]	0.00
Target-formation failure	0.63 [0.51–0.74]	0.14 [0.10–0.19]	0.22

C. Retained-cohort sensitivity to the repository scalar reference

Model	Outcome	Subjects	Windows	Aligned-target MAE	Repository-scalar MAE	Change
PhysioCAT	SBP	2,714	186,252	4.26	4.28	+0.02
PhysioCAT	DBP	2,714	186,252	3.11	3.12	+0.01
Matched no-delay	SBP	2,714	186,252	5.25	5.27	+0.02
Matched no-delay	DBP	2,714	186,252	3.51	3.53	+0.01
MuFuBP-Net	SBP	2,714	186,252	5.10	5.12	+0.02
MuFuBP-Net	DBP	2,714	186,252	3.38	3.39	+0.01

D. Reference-ABP quality sensitivity

Evaluation set	Windows	Median ABP SQI	PhysioCAT	Matched no-delay	MuFuBP-Net
All retained windows	186,252	0.67	4.26/3.11	5.25/3.51	5.10/3.38
ABP SQI ≥ 0.42	182,642	0.67	4.26/3.10	5.24/3.51	5.09/3.37
ABP SQI ≥ 0.55	154,531	0.70	4.24/3.06	5.18/3.47	5.02/3.33

E. Blinded reference-ABP waveform review

Retained ABP SQI median [IQR]	Below 0.42	Reviewed windows / subjects	Usable	Agreement / κ	ABP-SQI AUC	Sensitivity / specificity
0.67 [0.58–0.75]	3,610 (1.9%)	800 / 678	79.0%	0.866 / 0.608	0.907	0.875 / 0.720

The fixed-seed balanced analysis included 1,024 windows from 256 subjects (4 per subject), sampled from the PAT-detected outer-test windows.

Table S29: Descriptive association between expected attention delay and the offline apparent-PAT estimate. Observed intervals use subject-cluster bootstrap; null ranges summarize permutation or random-weight distributions.

Analysis	Metric	Estimate	95% interval / null range	Windows
Observed sparse attention	Pearson r: apparent PAT vs expected attention delay	0.586	[0.543, 0.626]	1,024
Within-subject centered association	Pearson r after subject-mean centering	0.605	[0.557, 0.651]	1,024
Window-level PAT permutation	Null Pearson r across 100 permutations	-0.002	[-0.057, 0.078]	1,024
Within-subject window permutation	Null centered Pearson r across 100 permutations	0.008	[-0.064, 0.081]	1,024
20 band-constrained random-weight seeds	Null Pearson r across random weights	-0.004	[-0.047, 0.057]	1,024